

College Student Sleep Habits Survey

Introduction

This study aims to identify correlations between daytime/nighttime sleep, caffeine intake, and self-reported academic stress across different student demographics.

Methodology

This is based on the Google form [Google form Link](#). The Google form is to ask College going students the following questions -

- 1) Name
- 2) Location
- 3) Name of Institute
- 4) Gender
- 5) Age
- 6) What is your current degree program?
- 7) What is your primary department or branch?
- 8) On a typical weeknight, how many hours of sleep do you get?
- 9) What time do you usually go to sleep at night?
- 10) What time do you sleep in the afternoon? (if you)
- 11) What time do you sleep any other duration of the day? (write NULL, if no)
- 12) On a scale of 1 to 10, how would you rate your average academic stress level this semester?
- 13) How many caffeinated beverages (coffee, tea, energy drinks) do you consume on an average day?

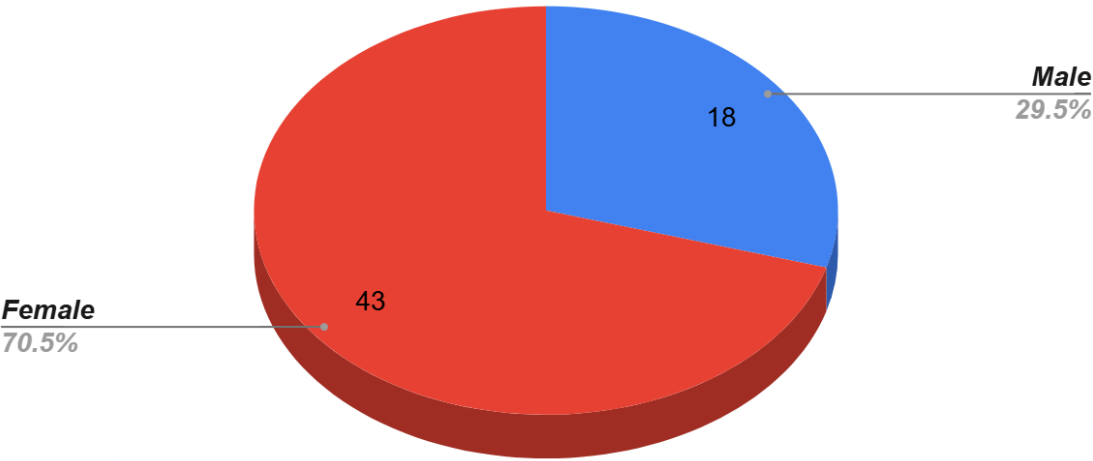
Table

Here are some Stats from the form

Gender Vs Count

Gender	Count
Female	43
Male	18
Grand Total	61

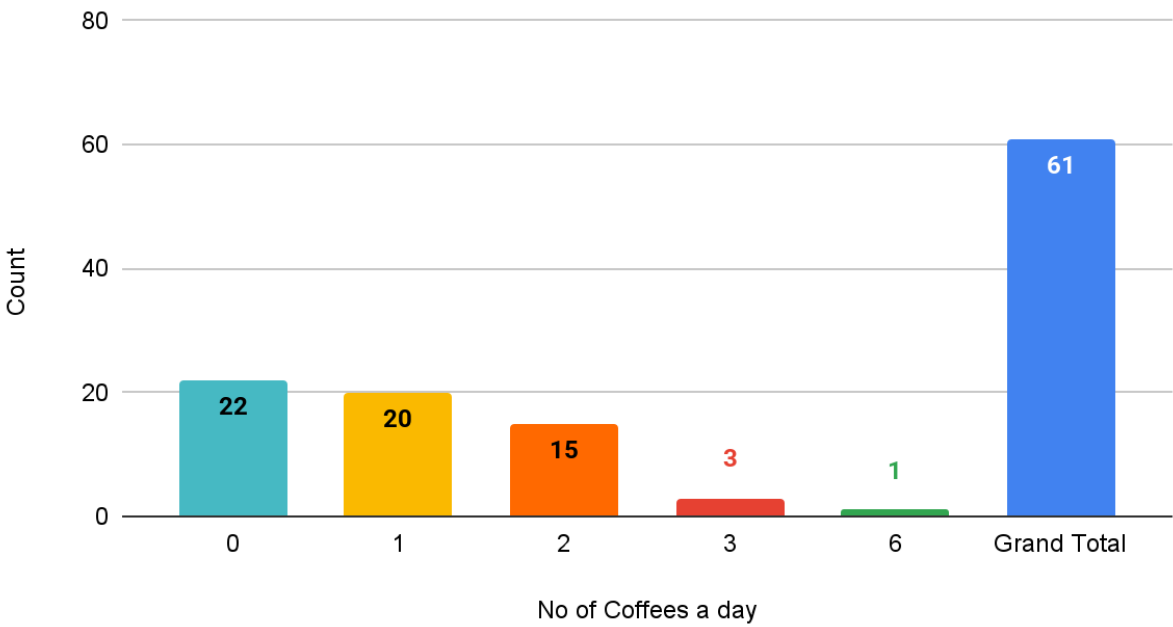
Count of Gender



Coffee Count

No of Coffees a day		Count
	0	22
	1	20
	2	15
	3	3
	6	1
Grand Total		61

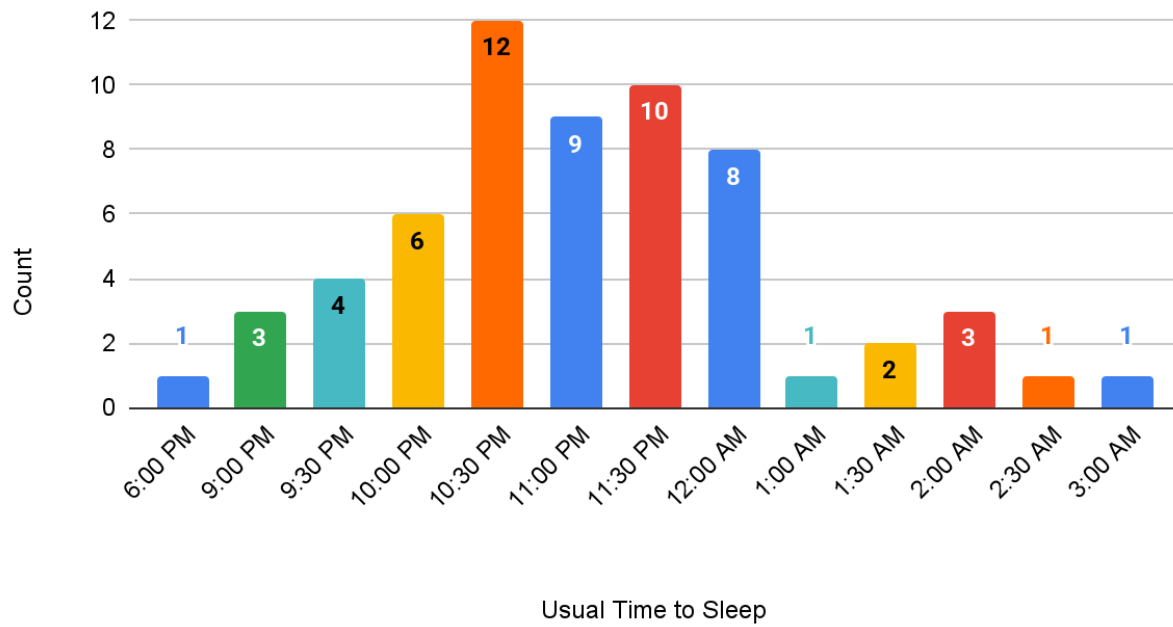
Count vs. No of Coffees a day



Usual Sleep Time

Usual Time to Sleep	Count
6:00 PM	1
9:00 PM	3
9:30 PM	4
10:00 PM	6
10:30 PM	12
11:00 PM	9
11:30 PM	10
12:00 AM (Midnight)	8
1:00 AM	1
1:30 AM	2
2:00 AM	3
2:30 AM	1
3:00 AM	1

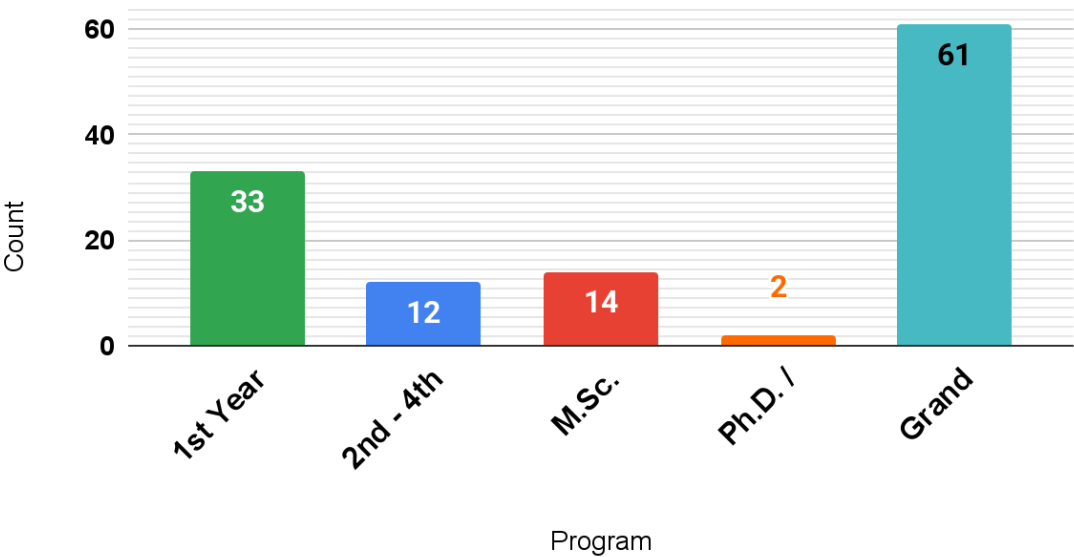
Count vs Usual Time to Sleep



Current Degree Program count

Program	Count
1st Year Undergraduate (B.Tech / B.S.)	33
2nd - 4th Year Undergraduate	12
M.Sc.	14
Ph.D. / Research Scholar	2
Grand Total	61

Count vs. Program

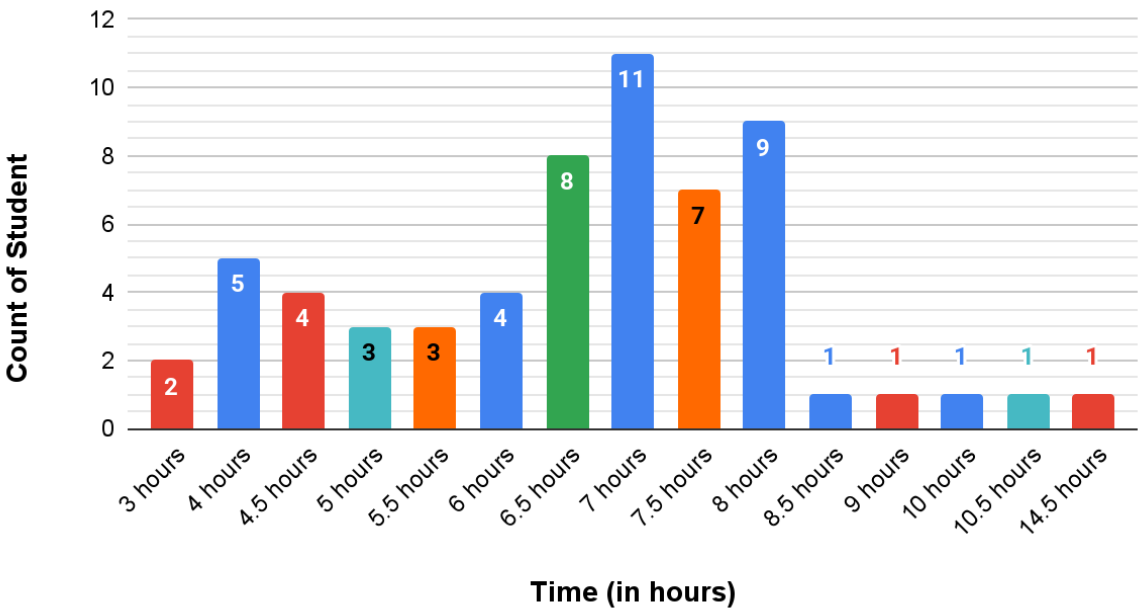


Number of Sleep hours Count

Time	Count
3 hours	2
4 hours	5
4.5 hours	4
5 hours	3
5.5 hours	3
6 hours	4
6.5 hours	8
7 hours	11
7.5 hours	7
8 hours	9
8.5 hours	1
9 hours	1
10 hours	1
10.5 hours	1
14.5 hours	1

Grand Total	61
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Count vs. Time

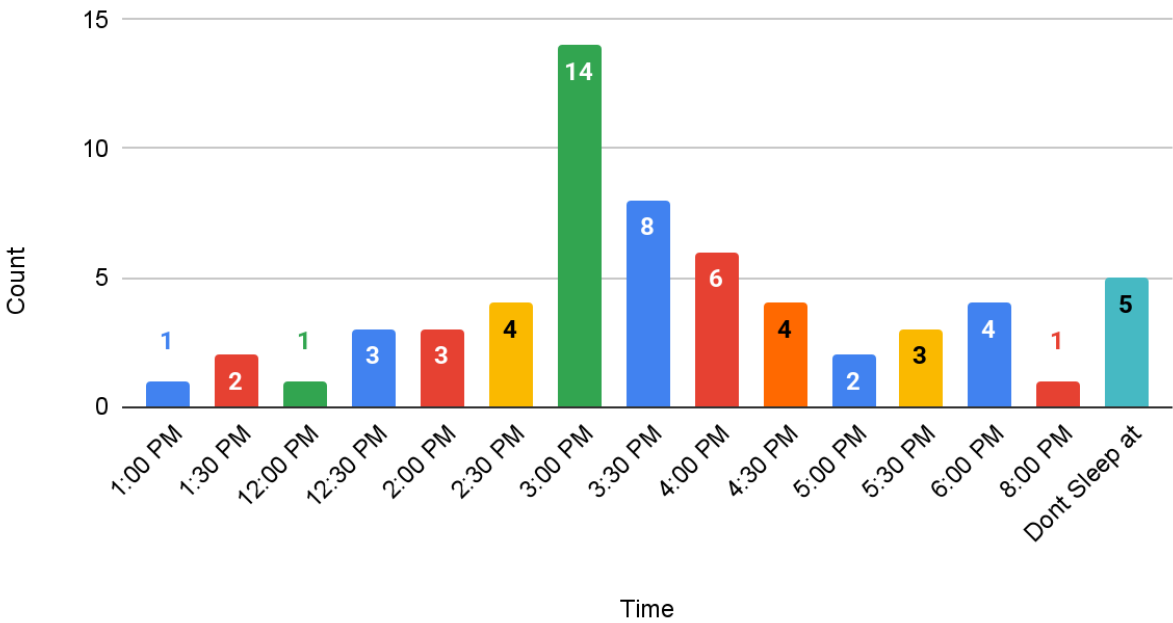


Afternoon Sleep time count

Time	Count
1:00 PM	1
1:30 PM	2
12:00 PM (Noon)	1
12:30 PM	3
2:00 PM	3
2:30 PM	4
3:00 PM	14
3:30 PM	8
4:00 PM	6
4:30 PM	4
5:00 PM	2
5:30 PM	3
6:00 PM	4

8:00 PM	1
Dont Sleep at all	5
Grand Total	61

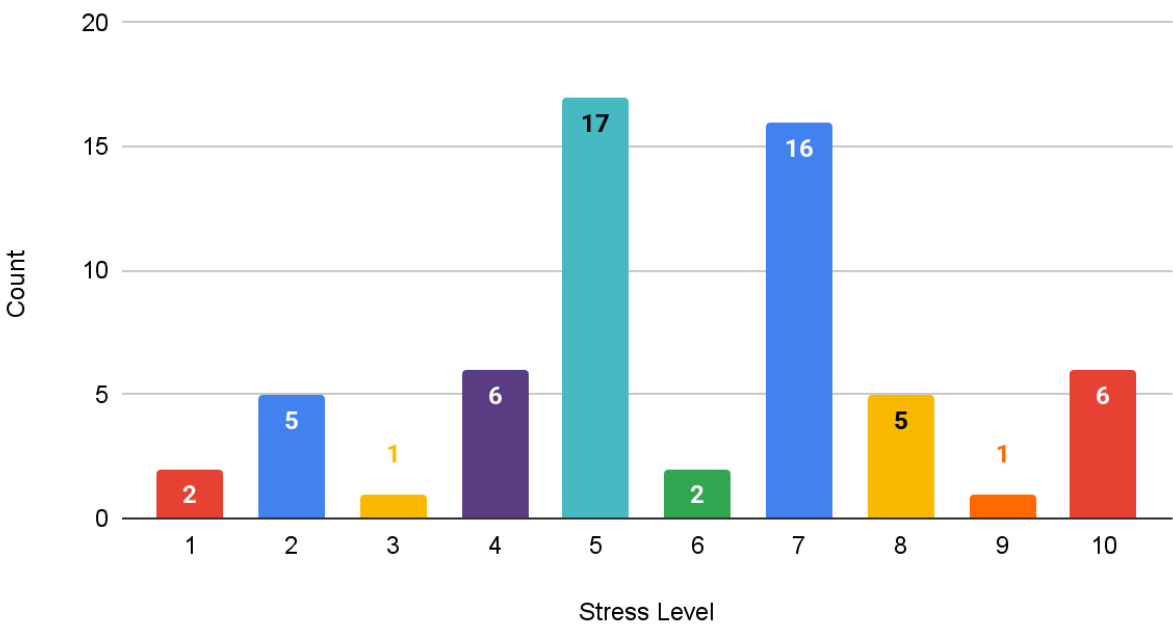
Count vs. Afternoon Sleeptime



Stress Level Count

Stress Level	Count
1	2
2	5
3	1
4	6
5	17
6	2
7	16
8	5
9	1
10	6
Grand Total	61

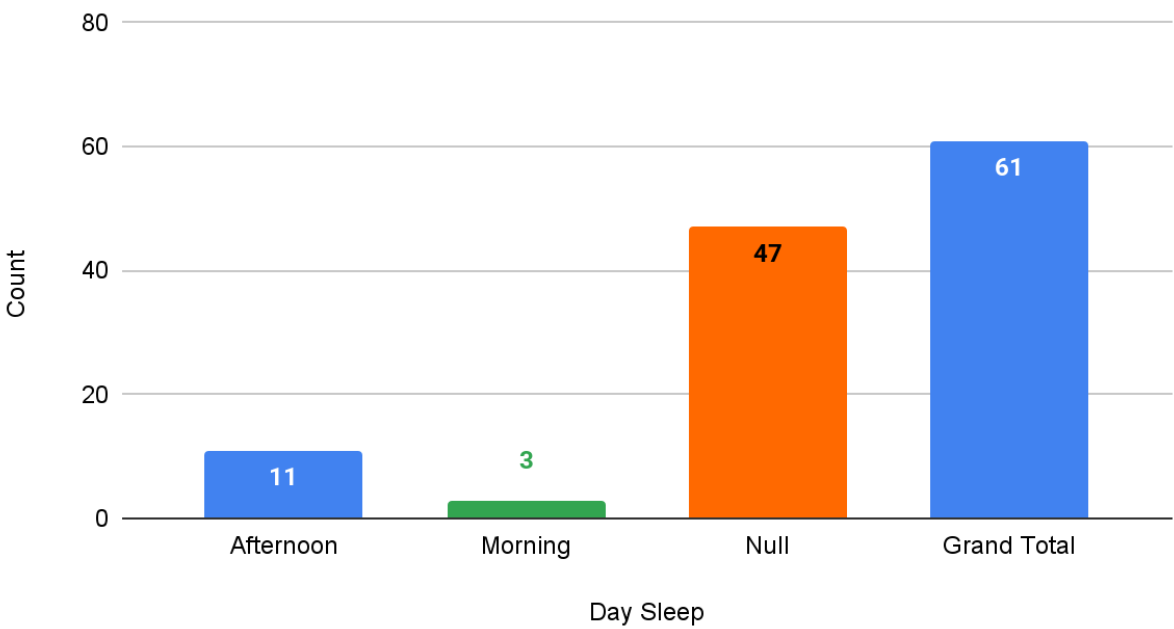
Count vs. Stress Level



Usual Day Sleep Phase Count

Day Sleep	Count
Afternoon	11
Morning	3
Null	47
Grand Total	61

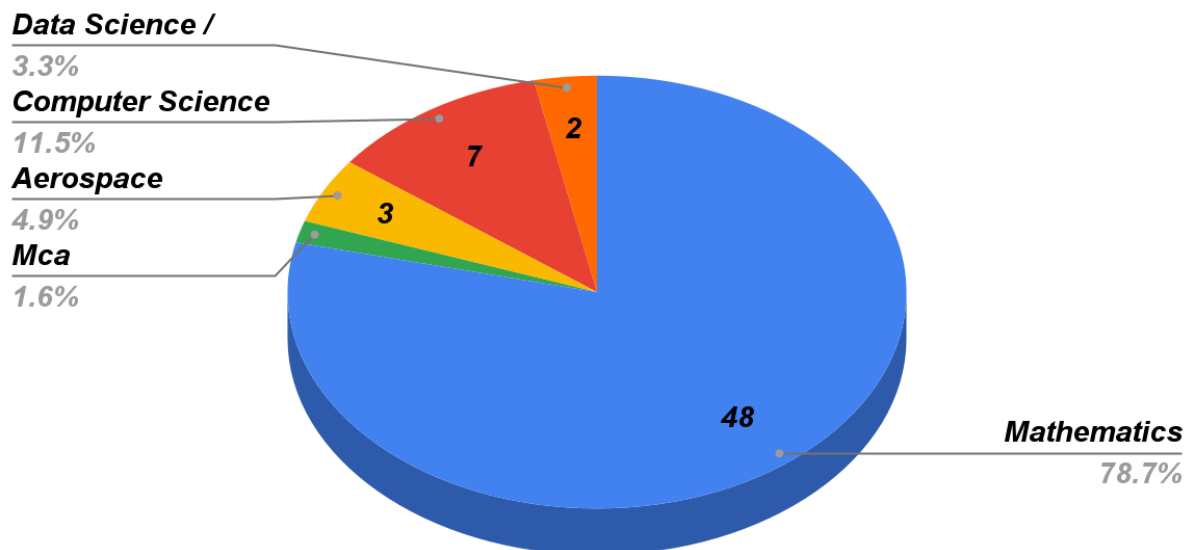
Count vs. Day Sleep



Department Count

Department	Count
Aerospace Engineering	3
Computer Science and Engineering (CSE)	7
Data Science / Artificial Intelligence	2
Mathematics	47
Mathematics	1
Mca	1
Grand Total	61

Count of Primary departments



Question answer

Now, as the data is not having 30 boys and girls each. I am unable to use z test thus, I am using 2 Table t test.

The following research questions explore the primary correlations identified in the data:

1. The Core Triangle: Stress, Sleep, and Caffeine

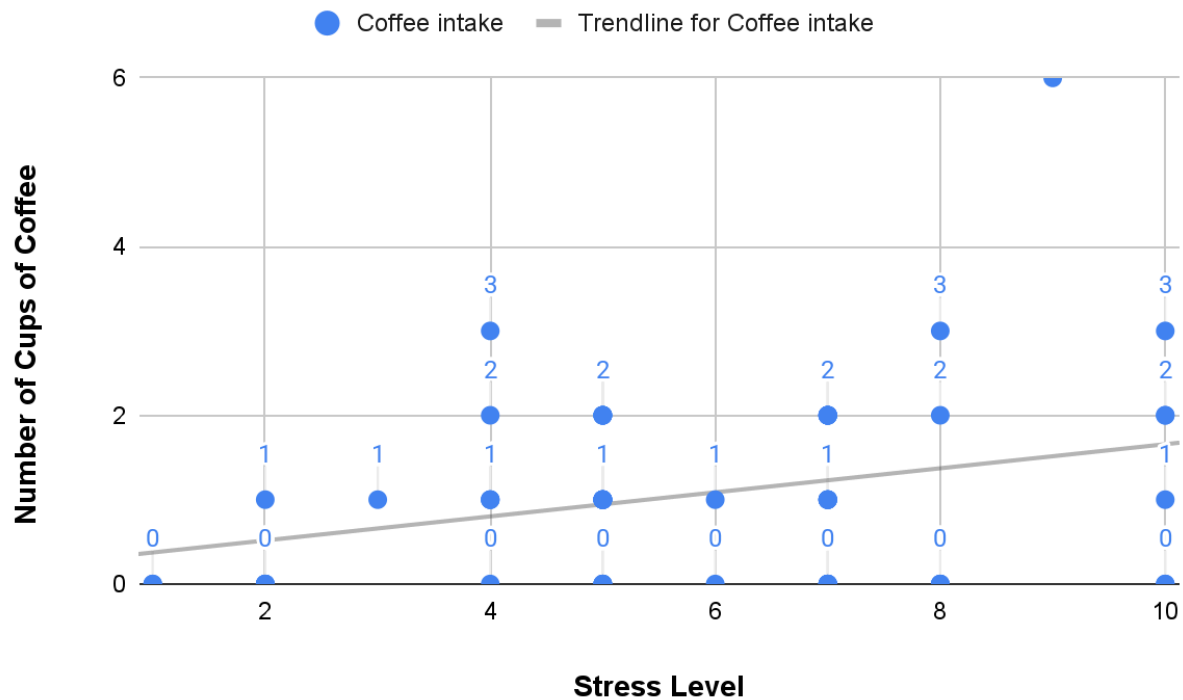
These questions examine the direct relationships between the most dynamic variables.

Question: Does higher caffeine intake correlate with higher academic stress?

Analysis: A weak positive correlation ($r = 0.297$) was observed between caffeine consumption and academic stress.

Methodology: A Scatter Plot (Caffeine on the X-axis, Stress on the Y-axis) with a trendline was used to visualize this relationship. The linear regression formula is expressed as $y - y' = byx(x - x')$.

Graph:

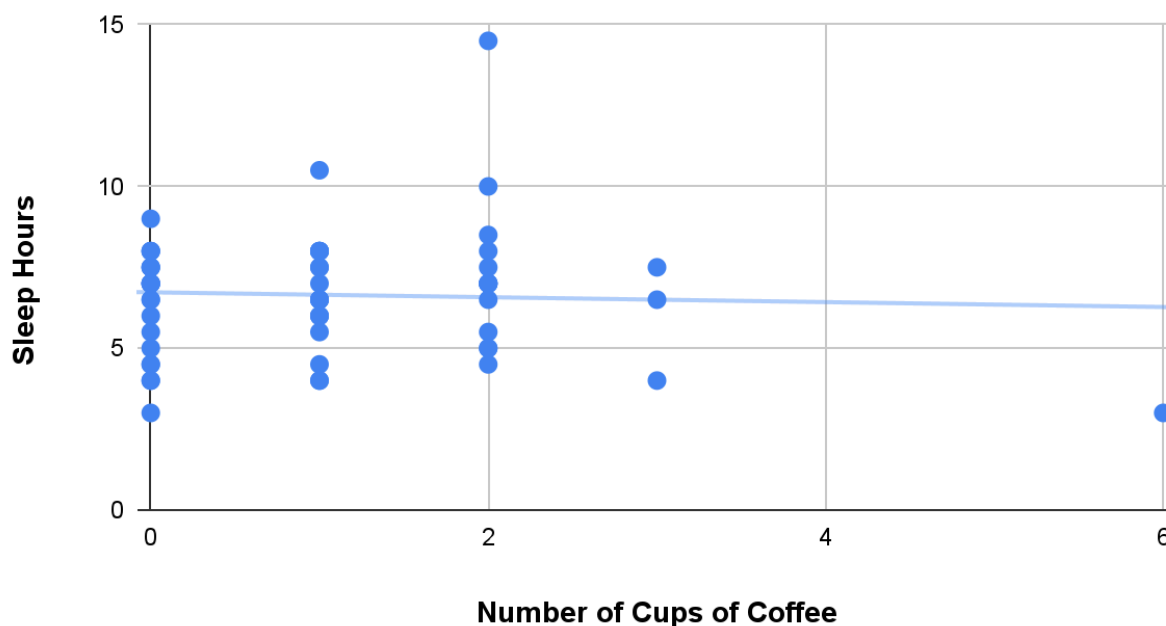


Based on the graph we can see a trend line.

Question: Do students who drink more caffeine sleep less at night?

Analysis: The data shows a negligible weak negative correlation ($r = -0.045$) between caffeine intake and nighttime sleep hours, despite the small sample size ($n=61$).

Coffee intake and Sleep Hours

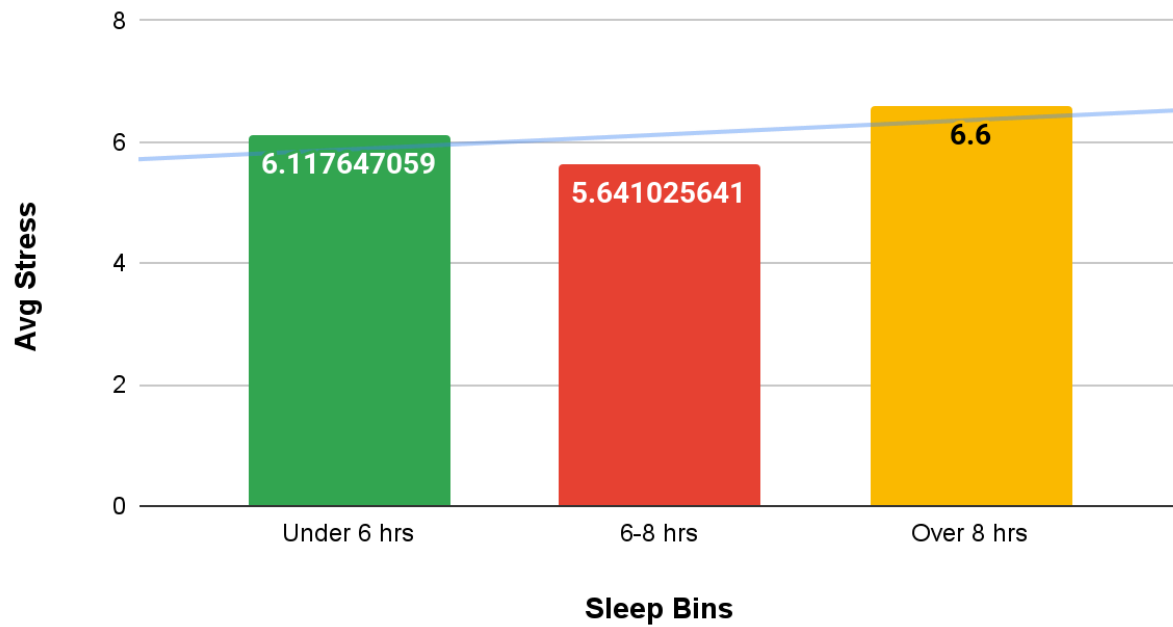


Question: Is there an optimal duration of sleep that correlates with minimized stress?

Analysis: A moderate positive correlation ($r = 0.613$) was found when comparing average stress levels against binned sleep duration groups, suggesting that sleep time is a significant factor.

Methodology: A Box Plot or Violin Plot was constructed (Sleep duration bins on the X-axis, Stress levels on the Y-axis). The correlation coefficient (r) was calculated using the average sleep time within each bin as the x-axis variable.

Avg Stress vs. Sleep Bins

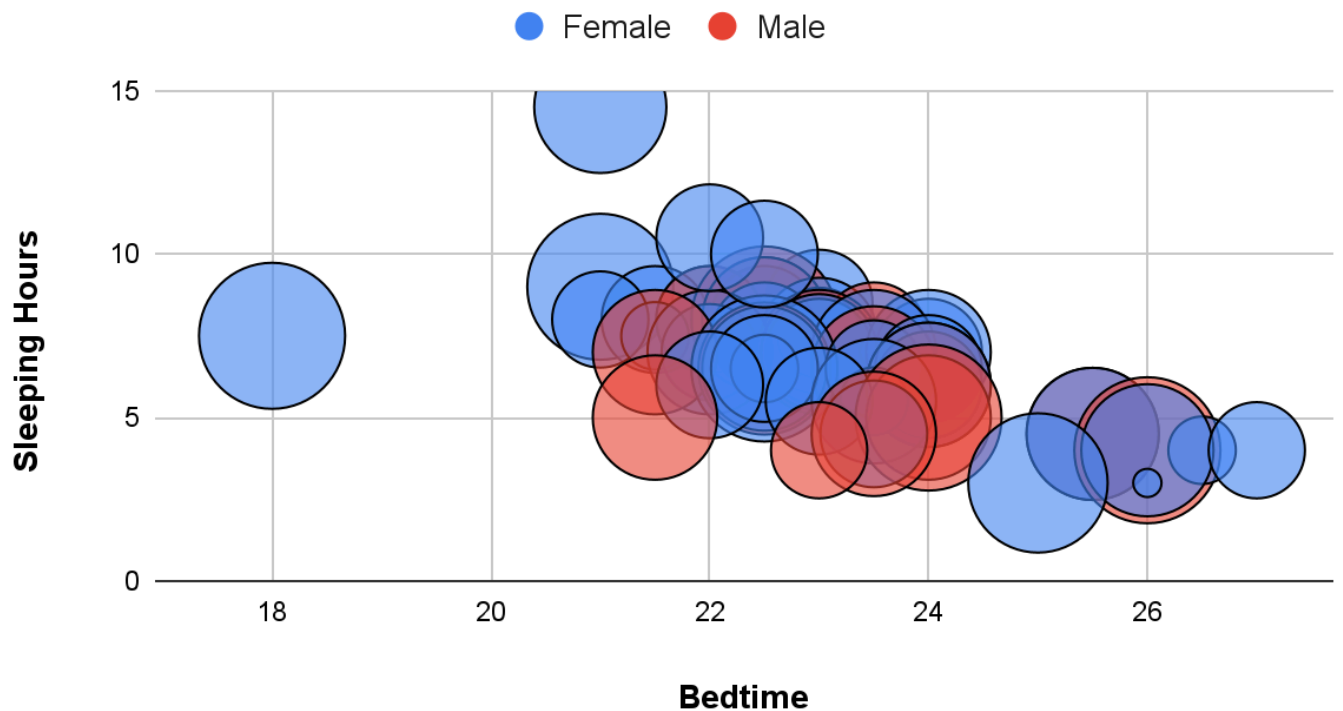


Question: Does a later bedtime correlate with reduced sleeping hours and elevated stress levels?

Methodology: A Bubble Chart was generated to integrate multiple variables:

- **X-axis:** Bedtime (Ascending from 6:00 PM)
- **Y-axis:** Sleeping Hours
- **Bubble Size:** Academic Stress Level
- **Bubble Color:** Gender (Blue = Male, Red = Female)

Bedtime Vs Stress



Objective Interpretation: The primary objective is to identify clusters where early bedtimes and average sleep hours correspond to smaller bubble sizes, indicating low stress levels.

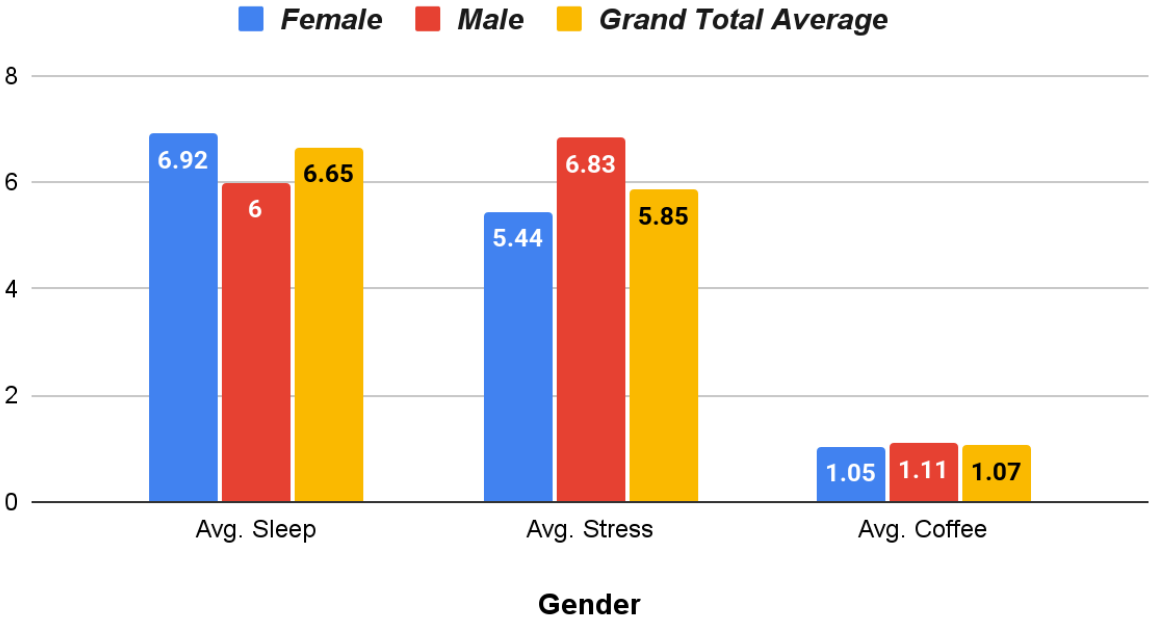
Section 2: Demographic Breakdowns

Question: How do stress, caffeine, and sleep habits differ across genders?

Methodology: Grouped Bar Charts showing side-by-side bars for Average Stress, Average Caffeine, and Average Sleep by gender.

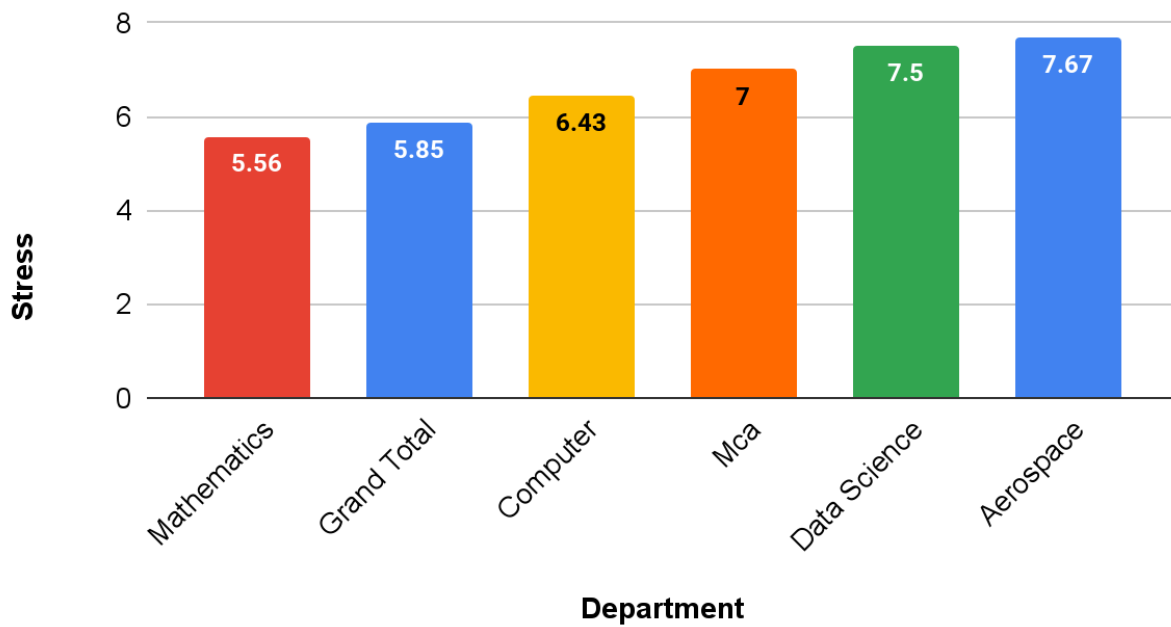
Graph:

Avg. Sleep, Avg. Stress and Avg. Coffee



Question: Which department experiences the highest average academic stress?
Methodology: Horizontal Bar Chart with departments sorted from highest to lowest average stress.
Graph:

Stress vs. Department

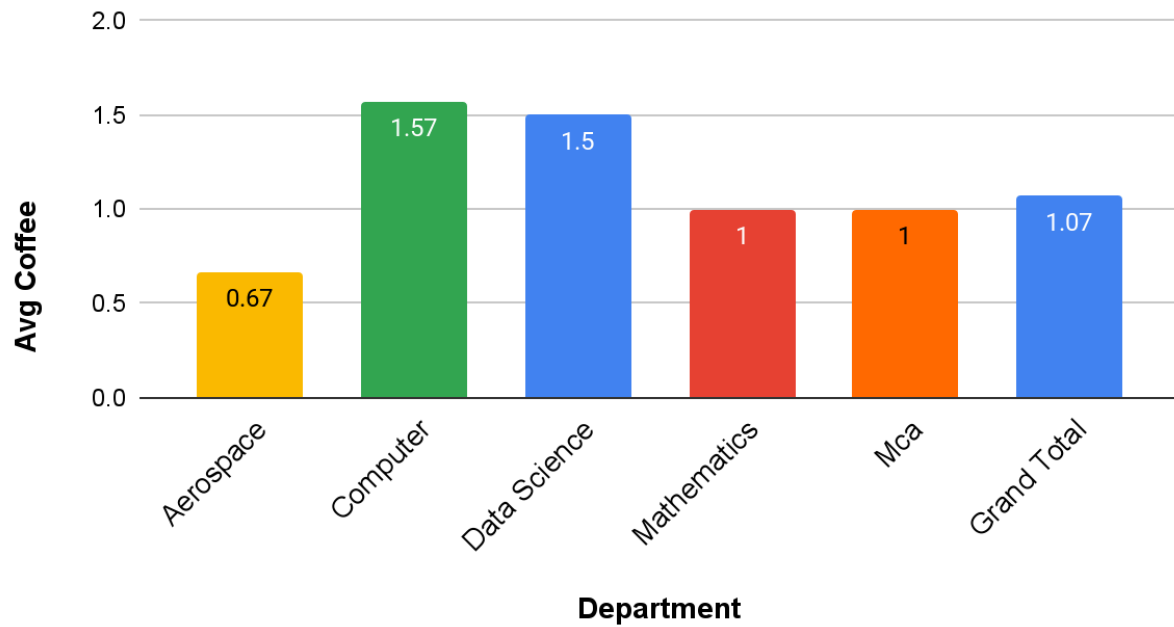


Question: Are certain departments more reliant on caffeine than others?

Methodology: Bar Chart or Box Plot comparing departments against the number of caffeinated drinks.

Graph:

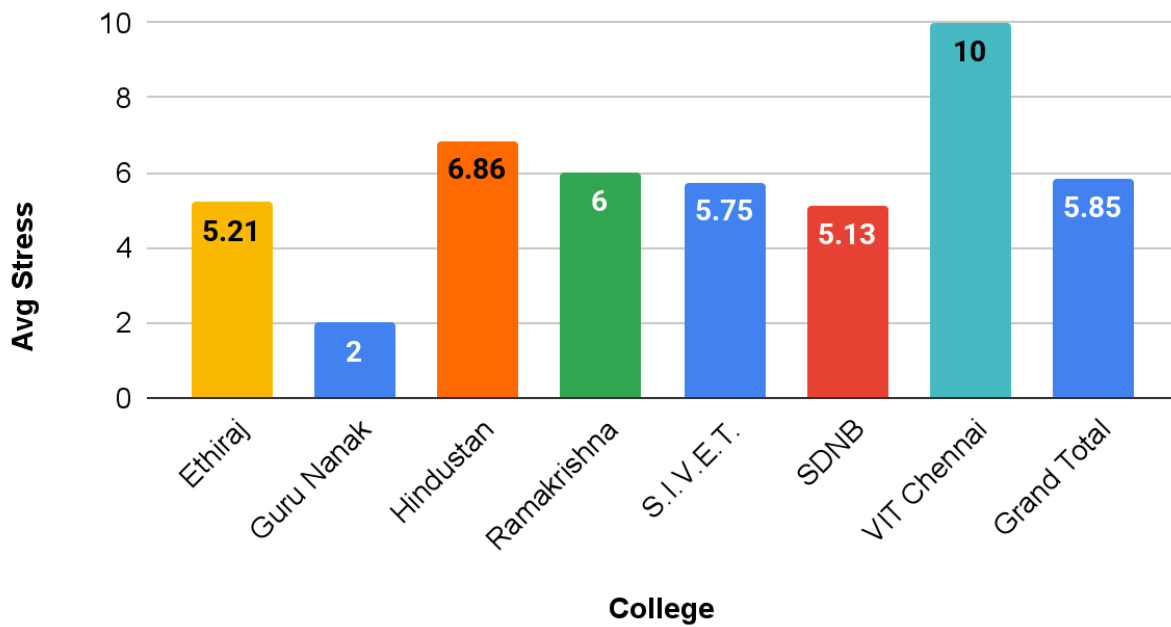
Department vs Avg Coffee



Question: Does the location or specific College Name impact stress levels?

Methodology: Map Visualization or a Bar Chart comparing average stress levels by college.
Graph:

Avg Stress vs. College



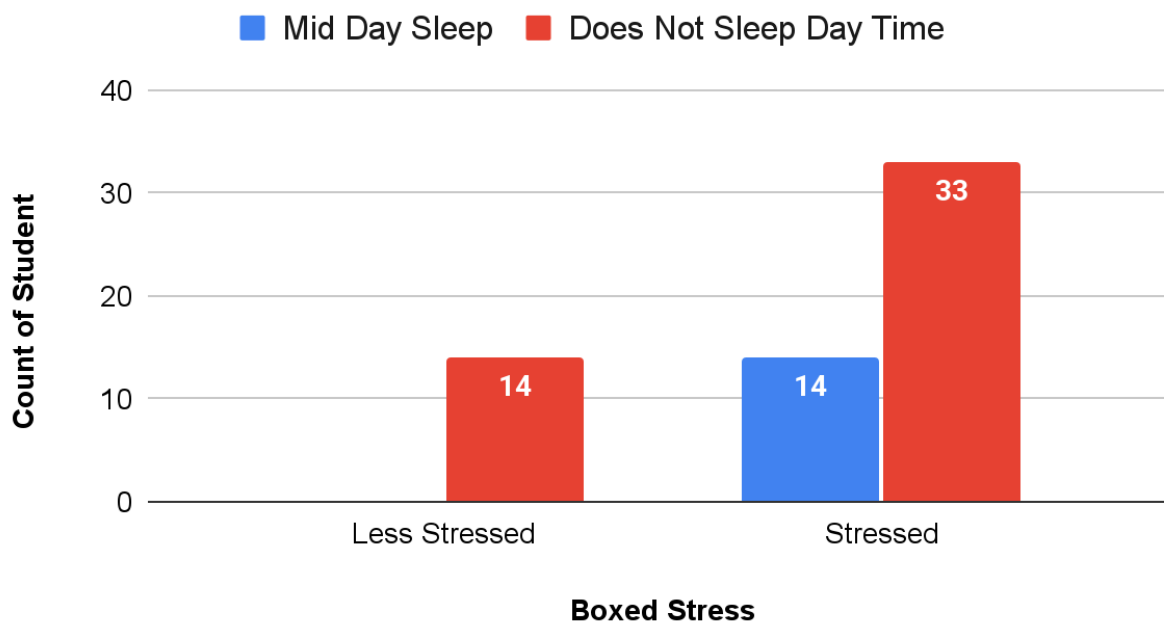
Section 3: The Napping Factor

Question: Does napping during the day lower academic stress?

Methodology: Side-by-Side Histograms or a Box Plot comparing stress levels of nappers versus non-nappers.

Graph:

Mid Day Sleep, Null and Grand Total



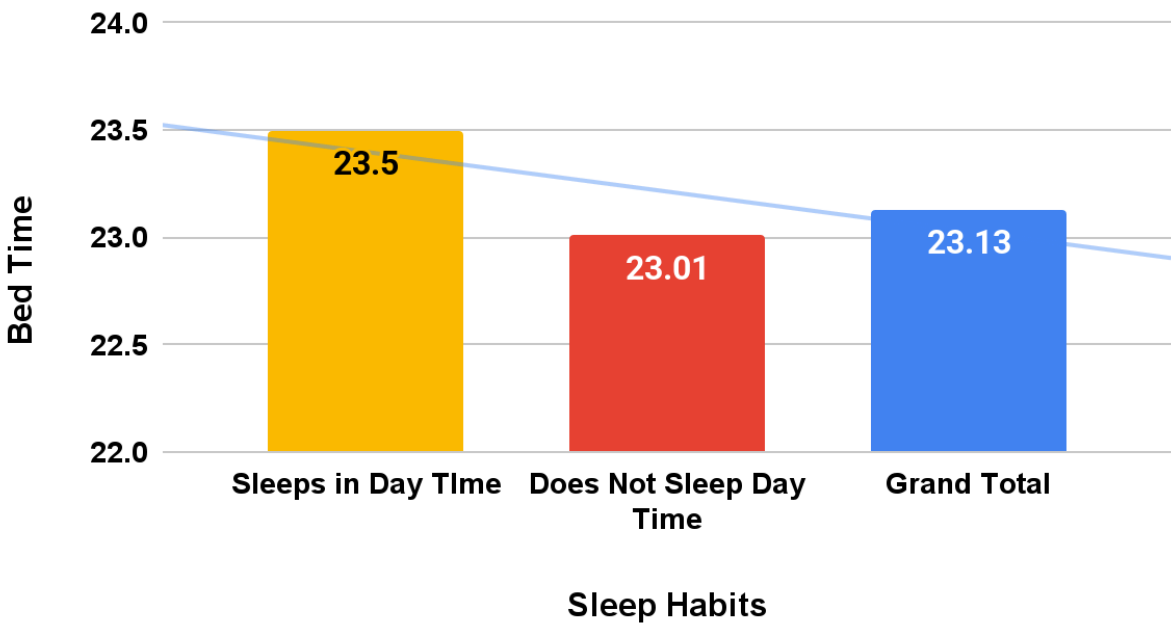
(Lower Stress is from 0-4 and Higher is 5-10)

Question: Is there a relationship between daytime napping and late bedtimes?

Methodology: Scatter Plot of bedtime versus daytime sleep duration.

Graph:

Bed Time vs. Sleep Habits



Section 4: Advanced Deep Dive Combinations

Question: Who are the high-functioning, high-stress students based on caffeine, sleep, and department?

Methodology: Heatmap with departments as rows, caffeine intake as columns, and color intensity representing average stress levels.

Table:

Department	0 Coffee's	1 Coffee	2 Coffee's	3 Coffee's	6 Coffee's	Grand Total
Aerospace Engineering	8		7			7.666666667
Computer Science and Engineering (CSE)	2	8	7	6		6.428571429
Data Science / Artificial Intelligence		5	10			7.5
Mathematics	5.315789474	4.8	6.25	10	9	5.5625

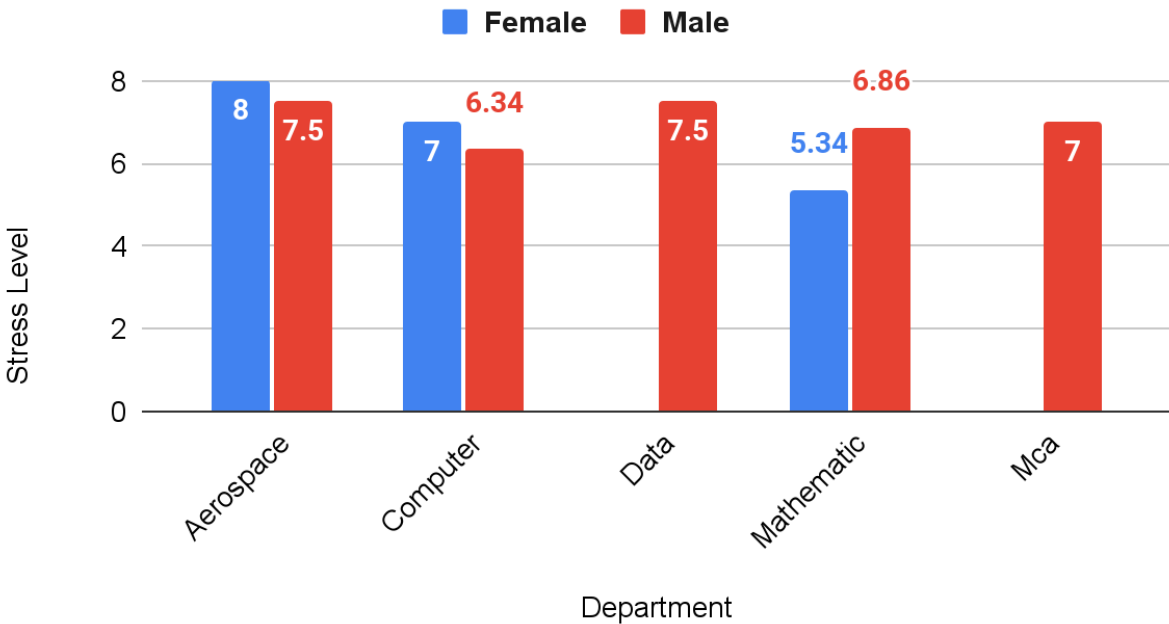
Mca		7				7
Grand Total	5.409090909	5.4	6.6	7.333333333 33	9	5.8524590 16

Question: Do different genders within the same department handle stress differently?

Methodology: Stacked or Grouped Bar Chart broken down by department and gender.

Graph:

Department Stress faced by Males and Females



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Statistical Methodology Adjustment

Since the male demographic consists of 18 participants—falling below the standard statistical threshold of 30 for large-sample tests—it is necessary to utilize a **Two-Sample T-test** rather than a Z-test. This adjustment ensures the findings remain statistically valid despite the smaller sample size.

1. Rationale for the T-test

A Z-test assumes a large, normally distributed population. When a group size is limited, a Z-test may become overly sensitive, potentially yielding a "false positive" or Type I error. The T-test provides a more conservative "safety net," ensuring that any identified differences are truly significant and not merely the result of sample variance.

2. Research Consistency

Despite this change in calculation, your research framework remains intact:

- **Hypotheses:** The Null (H_0) and Alternative (H_a) hypotheses regarding gender-based sleep differences remain unchanged.
- **Data Requirements:** You will continue to use the Mean and Standard Deviation for both demographics (41 female, 18 male participants).

3. Computational Execution

Modern spreadsheet software simplifies this process. If your raw data is organized in columns—for instance, Female sleep data in $A2:A42$ and Male sleep data in $B2:B19$ —you can execute the test with the following formula:

None

```
=T.TEST(A2:A42, B2:B19, 2, 3)
```

Parameters Defined:

- **2 (Two-tailed):** Used to detect any significant difference in either direction.
- **3 (Welch's T-test):** Specifically handles unequal sample sizes and variances, providing the most robust result.

4. Interpretation of Results

The formula will result in a **p-value**, which serves as your decision metric:

- **$p < 0.05$:** Reject the null hypothesis. There is a statistically significant difference in sleep habits between genders.
- **$p > 0.05$:** Fail to reject the null hypothesis. Any observed difference is likely due to chance.